Robotic Navigation HW3-2 Report

Deep Reinforcement Learning on Proly

114062633 / Shin-Cheng Huang

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1 Reward Function Design

The core of this experiment lies in the logical design of the reward function to guide the AI in finding optimal paths within complex terrains. The following details the design of each reward component:

1.1 Checkpoint Capture Reward

To provide a strong directional incentive for the AI, a significant positive reward of **+200.0** is granted when the robot successfully passes a checkpoint. This high value compensates for environmental penalties accumulated during the process, ensuring that "capturing the flag" remains the primary motivation throughout the agent's life cycle.

1.2 Continuous Distance Reward with Guard Logic

Instead of simple binary rewards (e.g., +1 for closer, -1 for farther), we implemented a smooth continuous shaping reward:

$$R_{dist} = (d_{prev} - d_{curr}) \times 5.0 \tag{1}$$

This design provides a fine-grained gradient for navigation.
Capture-frame Guard Mechanism: In practice, the moment a flag is captured, the target jumps to the next distant flag, causing d_{curr} to increase sharply and triggering a massive penalty. We implemented logic to detect changes in `last_checkpoint_index`; if a change is detected, the distance reward for that specific frame is set to zero. This significantly improves training stability around the checkpoint areas.

1.3 Time Penalty (Efficiency and Anti-idling)

To prevent the AI from idling or moving aimlessly due to fear of collisions, a constant time penalty of **-0.02** is subtracted per frame. This forces the robot to prioritize path efficiency rather than adopting a "wandering" or "idling" strategy.

1.4 Terrain Detection Penalty (Grid-based Avoidance)

Using the 5×5 terrain grid, the robot dynamically perceives obstacles. We designed a gradient penalty based on Chebyshev distance for water or mud pits (terrain index -1):

- **Collision/Occupancy (distance 0):** -5.0
- **Immediate Proximity (distance 1):** -2.0
- **Edge Perception (distance 2):** -1.0

This allows the AI to perceive "pain" before actual contact, enabling it to proactively navigate around hazardous terrain and increasing the overall success rate of the mission.